

# Section 1.2: Linear Functions and Applications

Finite Mathematics MTH 132 Spring 2014

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## Example 1.2.1.

Let  $g(x) = -4x + 2$ . Find  $g(2), g(-1), g(0), g(x+1)$ .

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- 5 Use your solution to answer *all* the questions.
- 6 Check!

# Cost Function

## Example 1.2.2.

The marginal cost of using a firm charges \$10 **per hour** plus a **flat rate** of \$50. Find the cost function,  $C(x)$ , of using this moving firm for  $x$  hours.

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## Example 1.2.3.

A moving firm charges \$10 **per hour** plus an additional **service charge** of \$50. Find the cost function,  $C(x)$ , of using this moving firm for  $x$  hours.

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## Problem 1.2.1. Find the cost function and identify all variables used.

The internet site "Music.com" charges a \$10 registration fee plus 99 cents per song.

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Example 1.2.6.

Find the profit function.

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## Example 1.2.7.

How many books must be produced and sold in order to break even?

# Revenue and Profit Function

A small publishing company has fixed costs of \$525, and can print a book for \$2. The same book sells for \$5 a piece.

## Example 1.2.8.

How many books must be produced and sold in order to make a profit of \$1,000?

# Revenue and Profit Function

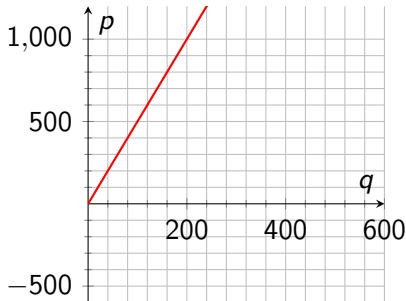
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Profit Function

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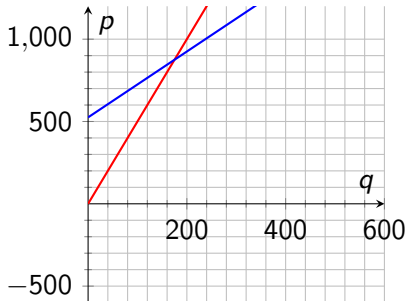
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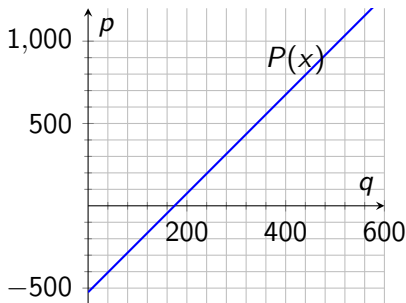
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- $D(q)$  is price at which consumers will buy  $q$  number of products.

## Example 1.2.10.

Suppose that the supply and price for strawberries are related by

$$p = S(q) = 0.25q$$

Find the price at the level of supply of 0 and 400 quarts

- $p$  is the price (in dollars).
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- $S(q)$  is the price at which suppliers will produce  $q$  number of products.

# Supply and Demand

## Example 1.2.11.

Find the *equilibrium quantity* for the supply,  $S(q) = 0.25q$ , and demand,  $D(q) = 5 - 0.25q$  functions.

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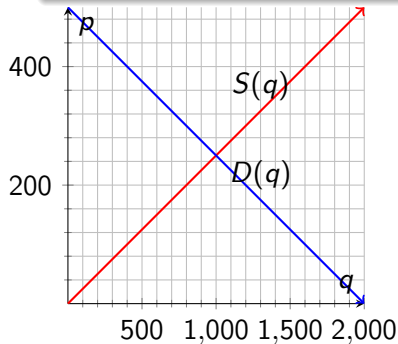
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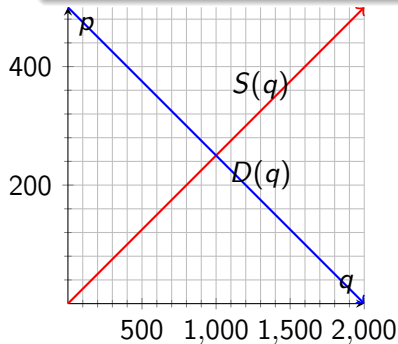
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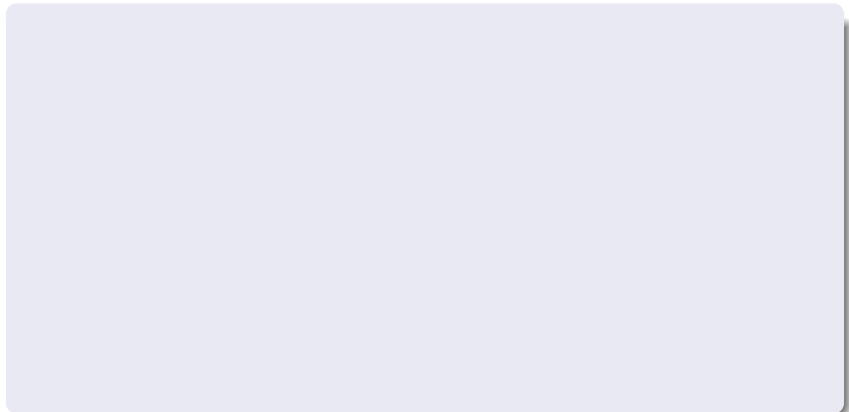


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